AI Hospital Information Chatbot

Project Brief

Problem Statement

Patients and visitors often need to search multiple websites or contact hospital staff to find information about departments, doctors, services, visiting hours, appointments, and hospital facilities. Patient feedback, public ratings, awards, and hospital-published success stories may also be scattered across different sources. This can make reliable information difficult to find and increase routine enquiries to hospital staff. The hospital needs a centralized chatbot that answers common questions using approved information and provides links to sources so users can verify public claims.

Proposed Solution

Develop a web-based AI chatbot that retrieves answers from the hospital's approved information and reliable public sources. It will provide concise answers, show relevant source links, and distinguish official hospital information from third-party ratings and reviews.

Features

- Answer questions about hospital location, contact details, departments, facilities, visiting hours, and services.
- Provide doctor profiles, specialties, qualifications, and approved clinic schedules.
- Share appointment booking links and relevant appointment guidance.
- Display public star ratings with the source name, retrieval date, and a link to the original review page when available.
- Present patient feedback from authorized or publicly accessible sources without inventing or altering reviews.
- Share hospital-published achievements, awards, and success stories with links to the original publications.
- Answer frequently asked questions about insurance, parking, visiting policies, and hospital procedures using approved information.
- Support Arabic and English when the selected model and knowledge sources support them.
- Use an approved knowledge base and retrieval-based responses to reduce unsupported answers.
- Escalate unanswered or sensitive enquiries to hospital staff.

Suggested Technology Stack

Frontend: React / Next.js **Backend:** FastAPI **Database:** PostgreSQL **AI:** an approved LLM with Retrieval-Augmented Generation (RAG) **Sources:** official hospital pages, approved documents, and permitted public review sources.

Trust, privacy and safety: Show ratings with their source and date; never invent ratings, testimonials, or successful outcomes. Share patient stories only with appropriate authorization. Protect personal and health information, and do not use the chatbot to diagnose patients or replace emergency services.